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MH1300 Foundations of Mathematics

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1. Definition of Invertibility

Definition | Invertible Matrix

Definition 1

A matrix $M \in \mathbb{M}_{m\times n}(\mathbb{R})$ is invertible iff:
1. M is a square matrix — i.e., $m = n$
2. There exists another matrix $M^{-1} \in \mathbb{M}_{n\times m}(\mathbb{R})$ such that $M^{-1}M = MM^{-1} = I_m$

There are other **characterizations** of invertibility, but the definition above is the fundamental definition. All other characterizations are simply consequences of this definition.

2. Exercises on Transpose

Example | Transpose Exercise

Example 2

Let A be an $n \times n$ square matrix. Let B be another $n \times n$ matrix whose (k, l)-entry is:

$$\sum_{p=1}^{n}(A_{pl}A_{pk} - A_{lp}A_{kp})$$

Which of the following is a general expression for B?

$$B = AA^T - A^TA$$

$$B = AA - A^TA^T$$

$$B = A^TA^T - AA$$

$$B = A^TA - AA^T$$

Solution Transpose Exercise

Solution 3

Notice that:

$$\begin{aligned}\sum_{p=1}^{n}(A_{pl}A_{pk}-A_{lp}A_{kp})= &\sum_{p=1}^{n}\Big(A_{pl}(A^T)_{kp}-(A^T)_{pl}A_{kp}\Big)\\ = &\sum_{p=1}^{n}\Big((A^T)_{kp}A_{pl}-A_{kp}(A^T)_{pl}\Big)\\ =&\sum_{p=1}^{n}\Big((A^T)_{kp}A_{pl}\Big)-\sum_{p=1}^{n}\Big(A_{kp}(A^T)_{pl}\Big)\end{aligned}$$

Therefore, we find that:

$$B=A^TA-AA^T$$

3. An Interesting Question

Example Product of Invertible Matrices

Example 4

Suppose A, B, C, D, E are matrices with inverses $A^{-1}, B^{-1}, C^{-1}, D^{-1}, E^{-1}$, respectively.

What is the inverse of $ABCDE$?

Solution Product of Invertible Matrices

Solution 5

BE CAREFUL! It is not $A^{-1}B^{-1}C^{-1}D^{-1}E^{-1}$.

The answer is:

$$E^{-1}D^{-1}C^{-1}B^{-1}A^{-1}$$

We can show this using the associativity of matrix multiplication:

$$\begin{aligned}(E^{-1}D^{-1}C^{-1}B^{-1}A^{-1})(ABCDE)=&(E^{-1}(D^{-1}(C^{-1}(B^{-1}(A^{-1}A)B)C)D)E)\\ = &(E^{-1}(D^{-1}(C^{-1}(B^{-1}(I)B)C)D)E)\\ = &(E^{-1}(D^{-1}(C^{-1}(B^{-1}B)C)D)E)\\ = &(E^{-1}(D^{-1}(C^{-1}(I)C)D)E)\\ = &\cdots\\ = &I\end{aligned}$$

Showing that it is a right-inverse is similar, and will not be shown here.

Thus, $E^{-1}D^{-1}C^{-1}B^{-1}A^{-1}$ is the inverse of $ABCDE$.

4. An Application of Inverse Matrices to Solving Systems of Equations

Theorem Solutions to Systems with Exactly One Solution — Theorem 6

We can express a system of equations as one matrix-vector multiplication:

$$A\vec{x} = \vec{b}$$

Suppose A were invertible. Then,

$$\vec{x} = A^{-1}\vec{b}$$

5. Organizing Matrices into Blocks

When performing matrix multiplication, you can subdivide matrices into blocks.

For instance:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{pmatrix} \to \begin{pmatrix} \begin{pmatrix} 1 & 2 \\ 5 & 6 \end{pmatrix} & \begin{pmatrix} 3 & 4 \\ 7 & 8 \end{pmatrix} \\ \begin{pmatrix} 9 & 10 \\ 13 & 14 \end{pmatrix} & \begin{pmatrix} 11 & 12 \\ 15 & 16 \end{pmatrix} \end{pmatrix}$$

Thus, we can express a matrix $\boldsymbol{A}$ with sub-blocks:

$$\boldsymbol{A} = \begin{pmatrix} \boldsymbol{A_{00}} & \boldsymbol{A_{01}} \\ \boldsymbol{A_{10}} & \boldsymbol{A_{11}} \end{pmatrix}$$

Multiplication of sub-blocks works similar to regular matrix multiplication. For instance:

$$\begin{pmatrix} \boldsymbol{A_{00}} & \boldsymbol{A_{01}} \\ \boldsymbol{A_{10}} & \boldsymbol{A_{11}} \end{pmatrix} \begin{pmatrix} \boldsymbol{B_{00}} & \boldsymbol{B_{01}} \\ \boldsymbol{B_{10}} & \boldsymbol{B_{11}} \end{pmatrix} = \begin{pmatrix} \boldsymbol{A_{00}B_{00} + A_{01}B_{10}} & \boldsymbol{A_{00}B_{01} + A_{01}B_{11}} \\ \boldsymbol{A_{10}B_{00} + A_{11}B_{10}} & \boldsymbol{A_{10}B_{01} + A_{11}B_{11}} \end{pmatrix}$$

However, **be careful**. Ensure that all multiplications between the sub-blocks are defined. That is, suppose you're multiplying two sub-blocks L and R:

$$LR$$

Make sure that the number of columns of L is equal to the number of rows of R.

6. A Theorem About the Solutions of a Linear System

Theorem Two Solutions Implies Infinite Solutions — Theorem 7

Suppose you have the following system:

$$\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= b_1 \\ \vdots \qquad\qquad & \quad \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

If the system has at least two distinct solutions, then it has infinitely many.

Proof Two Solutions Implies Infinite Solutions

Proof 8

Let $A = \left(a_{ij}\right)_{m\times n}$, $\vec{x} = (x_i)_{n\times 1}$, and $\vec{b} = (b_i)_{m\times 1}$. Then, the system is equivalent to the single matrix equation:

$$A\vec{x} = \vec{b}$$

Suppose $\vec{x_1}$ and $\vec{x_2}$ are two distinct solutions of the equation above.

Then,

$$A\vec{x_1} = \vec{b}$$

and

$$A\vec{x_2} = \vec{b}$$

Suppose $\lambda \in \mathbb{R}$.

Then, I claim that $\vec{x} = \lambda\vec{x_1} + (1-\lambda)\vec{x_2}$ are also solutions.

Notice that:

$$\lambda A\vec{x_1} + (1-\lambda)A\vec{x_2} = \lambda\vec{b} + (1-\lambda)\vec{b} \;\rightarrow\; A\lambda\vec{x_1} + A(1-\lambda)\vec{x_2} = \vec{b}$$

$$\rightarrow\; A(\lambda\vec{x_1} + (1-\lambda)\vec{x_2}) = \vec{b}$$

Therefore, $\lambda\vec{x_1} + (1-\lambda)\vec{x_2}$ is a solution of $A\vec{x} = \vec{b}$. Since every choice of $\lambda \in \mathbb{R}$ corresponds to some unique solution of $A\vec{x} = \vec{b}$, there are infinite solutions. Therefore, the existence of at least two solutions implies the existence of infinitely many, as needed. ■

Note: The expression $\lambda\vec{x_1} + (1-\lambda)\vec{x_2}$ is similar to **linear interpolation.** Geometrically, you can think of it as drawing a line between the two vectors.